

Multivariable Calculus
(MT2008)

Date: 21 September 2024

Course Instructor

Muhammad Yaseen

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

Roll No

Section

Student Signature

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Attempt all the questions.

CLO #1.

Q1: Use traces to sketch the quadric surface with equation

$$x^2 + 2z^2 - 6x - y + 10 = 0$$

[6 marks]

CLO #2.

Q2: (a). Determine whether the given function is continuous or discontinues.

$$g(x) = \begin{cases} (x^2 - y^2)/(x^2 + y^2) & \text{if } (x, y) \neq 0 \\ 0 & \text{if } (x, y) = 0 \end{cases} \quad [4 \text{ marks}]$$

(b). The total resistance R produced by three conductors with resistances R_1, R_2, R_3 connected in a parallel electrical circuit is given by the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad [4 \text{ marks}]$$

Find $\partial R / \partial R_1$.

(c). If $z = f(r, \theta)$, where $r = x \cos y$ and $\theta = x \sin y$, show that

$$\frac{\partial^2 z}{\partial r^2} + \frac{\partial^2 z}{\partial \theta^2} = \frac{\partial^2 z}{\partial x^2} + \frac{1}{x^2} \frac{\partial^2 z}{\partial y^2} + \frac{1}{x} \frac{\partial z}{\partial x}$$

[8 marks]

CLO #3.

Q3: Find the curvature of the twisted cubic $r(t) = \langle t, t^2, t^3 \rangle$ at point $(0, 0, 0)$. [8 marks]